RAJALAKSHMI ENGINEERING COLLEGE

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N2

Started on	Wednesday, 19 August 2026, 8:31 AM
State	Finished
Completed on	Wednesday, 19 August 2026, 8:36 AM
Time taken	4 mins 16 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int binarySearch(int arr[], int low, int high, int target) {
    if (low > high)
        return -1;

    int mid = (low + high) / 2;

    if (arr[mid] == target)
        return mid;
    else if (arr[mid] < target)
        return binarySearch(arr, mid + 1, high, target);
    else
        return binarySearch(arr, low, mid - 1, target);
}

int findPair(int arr[], int low, int high, int x, int *a, int *b) {
    if (low > high)
        return 0;

    int mid = (low + high) / 2;
    int target = x - arr[mid];

    int index = binarySearch(arr, mid + 1, high, target);

    if (index != -1) {
        *a = arr[mid];
        *b = arr[index];
        return 1;
    }

    if (findPair(arr, low, mid - 1, x, a, b))
        return 1;

    return findPair(arr, mid + 1, high, x, a, b);
}

int main() {
    int n, x;

    scanf("%d", &n);

    int arr[n];

    for (int i = 0; i < n; i++)
        scanf("%d", &arr[i]);

    scanf("%d", &x);

    int a, b;
```

```
51
52    if (findPair(arr, 0, n - 1, x, &a, &b)) {
```

	Input	Expected	Got	
✔	4 2 4 8 10 14	4 10	4 10	✔
✔	5 2 4 6 8 10 100	No	No	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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